Multi-Source Data Scraping & Trust Scoring System

1. Scraping Strategy

This project implements a multi-source scraping pipeline that collects structured data from three platforms: blogs, YouTube, and PubMed.

For blog scraping, HTTP requests are sent using the requests library with a browser-like header. HTML content is parsed using BeautifulSoup, and non-relevant elements such as ads, scripts, and navigation bars are removed. The main article content is extracted using prioritized selectors, and metadata (author, publication date) is retrieved from Open Graph tags and JSON-LD. Dates are validated to prevent future values, and language is detected using langdetect.

For YouTube, metadata such as title, channel name, and publish date is extracted using yt-dlp. Transcripts are fetched using youtube-transcript-api, with fallback to video descriptions if transcripts are unavailable. Logging is optimized to suppress non-critical warnings.

For PubMed, the NCBI E-utilities API is used to fetch structured XML data. The system extracts title, authors, journal, publication date, and abstract. Citation counts are approximated via the elink API, and content is split into smaller chunks for processing.

2. Topic Tagging Method

Topic tagging is implemented using a hybrid approach:

- **Taxonomy Matching:** A predefined keyword dictionary maps content to categories such as AI, Machine Learning, Healthcare, and Web Scraping using regex matching.
- **Keyword Extraction:** A frequency-based scoring method extracts meaningful bigrams and keywords (e.g., “Neural Network”, “HTTP Requests”), prioritizing longer and more informative terms.

The final output combines both approaches and limits the number of tags to maintain relevance.

3. Trust Score Algorithm

The Trust Score evaluates content reliability using a weighted combination of five factors:

Trust Score =
0.25 × author_credibility +
0.25 × domain_authority +
0.20 × recency +
0.20 × citation_count +
0.10 × medical_disclaimer

- **Author Credibility:** Based on known organizations, real-name patterns, and PubMed authorship.
- **Domain Authority:** Uses a curated domain map and TLD-based heuristics.

- **Recency:** Modeled using exponential decay to prioritize recent content.
- **Citation Count:** Log-normalized; scholarly sources benefit more.
- **Medical Disclaimer:** Penalizes health content lacking disclaimers.

Each record also includes a detailed score breakdown for transparency.

4. Edge Case Handling

The system handles common real-world issues:

- Missing author → low credibility score
- Missing date → recency penalty
- Missing transcript → fallback to description
- Multiple authors → averaged credibility
- Non-English content → detected and stored
- Future dates → flagged and penalized

5. Abuse Prevention Logic

To prevent manipulation, the system includes:

- Detection of fake or suspicious author names
- Penalization of low-quality or spam domains
- Strong penalties for misleading medical content
- Reduced scores for missing disclaimers
- Recency decay to avoid outdated information

6. Limitations

- No rate limiting for scraping requests
- Limited support for JavaScript-heavy websites
- Author verification is heuristic-based
- Citation counts are limited to PubMed data
- Topic tagging currently supports only English

Conclusion

This project demonstrates a complete pipeline for extracting, structuring, and evaluating content from multiple sources. The combination of robust scraping, intelligent tagging, and transparent trust scoring provides a scalable approach to assessing content reliability.